# Reflexes

# Objectives

1- Defining the importance of reflexes assessment in children.

2- Demonstrate how to assess reflexes in infants.

3- Discuss the type of reflexes( sucking, rooting, Babinski, startle and Moro reflexes …etc.).

**4-** Explain the child's response to each reflex according to the offered type .

# Reflexes

- Testing reflexes is an important part of the neurologic examination.
- Reflexes should be tested at the end of the examination because they may disturb the infant and interfere with auscultation.
- Reflexes are present at birth and disappear by 9 months of age.

# Classification of reflexes

- **Primitive reflexes:**

**A- feeding reflexes: sucking, rooting, and swallowing reflex**

**B- Motor reflex: like Moro , startle , palmer, planter,…etc.**

- Protective reflexes: Gag, cough, and sneeze reflexes also are present in newborns to maintain a clear airway.

**Primitive reflexes** originate in the central nervous system and are exhibited in normal infants

- **Rooting reflex:** The infant turns his head toward the mother's breast and begins to suck when his lower lip or cheek is stroked. should disappear at about age 3-4 months but may persist for up to 12 months.





- **Sucking reflex:** The infant moves his lips, mouth, and tongue and shows interest to feed. Present at birth; disappears at 10–12 months.



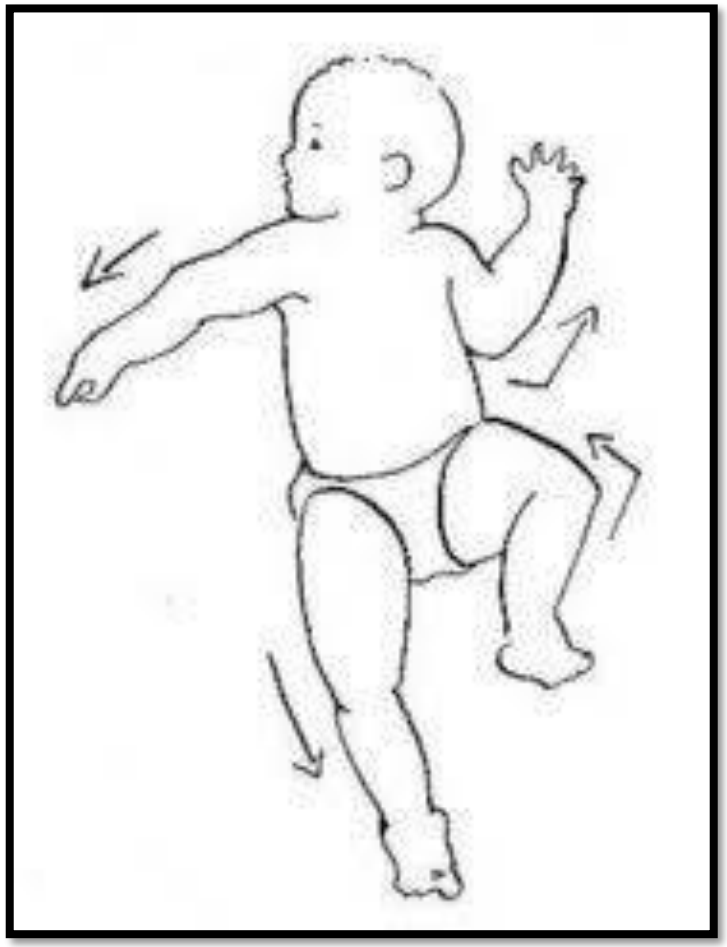


- **Startle reflexes:** Sudden loud noise causes abduction of arms with flexion of elbows; hands remain clenched; disappears by age 4 months





- **Tonic neck reflex** (or fencing posture): The infant turns his head to one side, and the arm on that side straightens while the opposite arm bends in a fencing position.





- **Moro reflex:** Sudden jarring or change in equilibrium causes sudden extension and abduction of extremities and fanning of fingers, with index finger and thumb forming a C shape, followed by flexion and adduction of extremities; legs may weakly flex; infant may cry; disappears after age 3–4 months.



- **Grasping reflex**: An object is placed in the infant's hand, and the infant's fingers close to grasp the object. palmar grasp lessens after age 3 months, to be replaced by voluntary movement; plantar grasp lessens by 8 months of age





- **Babinski reflex:** The sole of the foot is stroked, and the infant responds by turning the foot inward while his toes flare outward or flexion.



- **Dancing (or stepping) reflex:** If infant is held so that sole of foot touches hard surface, there is reciprocal flexion and extension of leg, simulating walking disappears after age 3–4 weeks, to be replaced by deliberate movement



- **Palmar grasp:** The examiner places a finger in the palm of the neonate's hand, he neonate grasps the finger tightly; if the neonate grasps the examiner's finger with both hands, the neonate can be pulled to a sitting position.
- **Plantar grasp:** Place a thumb firmly against the ball of the infant's foot, Toes flex tightly down in a grasping motion.






Thank you